Module 8 Vocabulary

3-Dimensional Figure: A solid shape that has length, width, and height, taking up space in three dimensions.

Altitude: A perpendicular segment from a vertex of a shape to the opposite side creating a right angle with the side.

Apex: The highest point of a cone or pyramid, where the lateral faces or curved surface meet.

Apothem: A line segment from the center of a regular polygon perpendicular to one of its sides.

Arc: A curved portion of a circle's circumference between two points on the circle.

Arc Length: The measure of the distance along an arc, found using a portion of the circle's circumference.

Area: The measure of the space enclosed within a two-dimensional shape, expressed in square units.

Area of a Sector: The portion of a circle's total area enclosed by two radii and an arc, found using a fraction of the circle's total area; $A = \frac{\theta}{360^\circ}\pi r2$.

Base: The bottom side of a geometric figure used as a reference for measurement. or The face of a 3D figure that serves as a foundation and is used to determine surface area and volume.

Center: The point in a circle that is equidistant from all points on the circle.

Central Angle: An angle whose vertex is at the center of a circle and whose sides extend to the circumference.

Chord: A line segment with both endpoints on the circle but does not necessarily pass through the center.

Circle: A set of all points in a plane that are equidistant from a given point called the center.

Circumference: The total distance around a circle, found using the formula $C = 2\pi r$.

Composite Figure: A shape made up of two or more basic geometric figures.

Cone: A 3D shape with a circular base and a curved surface that tapers to a single point called the apex.

Cylinder: A 3D shape with two parallel circular bases connected by a curved surface.

Degree: A unit of measurement for angles, where a full circle measures $360°$.

Diameter: A straight line passing through the center of a circle and touching both sides of the circumference, equal to twice the radius.

Face: A flat surface of a 3D figure, such as the bases and lateral faces of prisms and pyramids.

Height: The perpendicular distance from the base to the highest point of a shape.

Heron's Formula: A formula used to find the area of a triangle when the lengths of all three sides are known: $A = \sqrt{s(s-a)(s-b)(s-c)}$ where s is the semi-perimeter and a, b, and c are the side lengths of the triangle.

Hexagon: A polygon with six sides and six angles.

Intercepted Arc: The arc that lies between the two sides of an angle, often measured by the angle's degree.

Quadrilateral: A four-sided polygon with four angles.

Lateral Area: The sum of the areas of only the lateral faces (not including the base) of a 3D shape.

Lateral Face: A face of a 3D figure that is not a base, typically found in prisms and pyramids.

Major Arc: An arc that measures more than $180°$.

Minor Arc: An arc that measures less than $180°$.

Net: A two-dimensional pattern that can be folded to form a 3D shape.

Octagon: A polygon with eight sides and eight angles.

Parallelogram: A quadrilateral with opposite sides that are parallel and equal in length.

Pentagon: A polygon with five sides and five angles.

Perimeter: The total distance around the outside of a polygon. or The total length of the boundary of a 2D shape, used in some surface area calculations.

Pi (π): A mathematical constant approximately equal to 3.14159, representing the ratio of a circle's circumference to its diameter.

Polygon: A closed two-dimensional shape formed by three or more straight sides.

Prism: A 3D figure with two parallel, congruent bases and rectangular lateral faces.

Proportion: A mathematical statement that two ratios are equal, often used to solve for missing values.

Pyramid: A 3D figure with a polygonal base and triangular lateral faces that meet at an apex.

Radius: A line segment from the center of a circle to any point on the circle's circumference.

Rectangle: A quadrilateral with four right angles and opposite sides that are equal in length.

Rhombus: A parallelogram with all four sides of equal length.

Right Cylinder: A cylinder whose sides are perpendicular to its circular bases.

Right Prism: A prism whose lateral faces are perpendicular to its bases.

Sector: A region of a circle enclosed by two radii and an arc.

Segment of a Circle: The region of a circle bounded by a chord and the arc between the chord's endpoints.

Semi-Perimeter: Half of a polygon's perimeter, often used in Heron's formula.

Slant Height: The distance from the apex of a pyramid or cone to the edge of the base, measured along the lateral surface.

Sphere: A perfectly round 3D shape where all points are equidistant from the center.

Square: A quadrilateral with four equal sides and four right angles.

Surface Area: The total area of all faces or curved surfaces of a 3D figure.

Total Surface Area: The sum of the lateral area and the area of all bases in a 3D figure.

Trapezoid: A quadrilateral with exactly one pair of parallel sides.

Triangle: A polygon with three sides and three angles.

Vertex: A point where edges or faces of a 3D figure meet.

Volume: The amount of space a 3D shape occupies, measured in cubic units.